Zero Anaphora Resolution in Chinese and Its Application in Chinese-English Machine Translation

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Abstract. In this paper, we propose a learning classifier based on maximum entropy (ME) for resolving ZA in Chinese. Besides regular grammatical, lexical, positional and semantic features, we develop two innovative Web-based features for extracting additional semantic information of ZA from the Web. Our study shows the Web as a knowledge source can be incorporated effectively in the learning framework and significantly improves its performance. In the application of ZA resolution in MT, it is viewed as a pre-processing module that is detachable and MT-independent. The experiment results demonstrate a significant improvement on BLEU/NIST scores after the ZA resolution is employed.

Keywords: zero anaphora resolution, Web-based features, ME-based classifier, machine translation.

1 Introduction

In many natural languages, grammatical components that can be understood contextually by a reader are frequently unexpressed for discourse fluency. This is especially the case in many Asian languages [1], such as Chinese, Japanese and Korean, where a kind of anaphoric expression is frequently eliminated. This phenomenon is called zero-anaphora (ZA) in Natural Language Processing (NLP) [2].

For example, in sentence (1)[1], ϕ_1 and ϕ_2 in the subject position refer to the subject "姐姐 (my sister)" and the object "苹果 (apples) in their previous sentences respectively, and they can be recovered as "她 (she)" and "它们 (they)". ϕ_3 in sentence (2) refers to the object of the preposition phrase, "席子 (the matting)", and it can be recovered as "它 (it)". In sentence (3), the grammatical role of ϕ_4 is the object. It refers to "问题 (the problem)".

(1) [姐姐]$_1$来看我, ϕ_1给了我几个[苹果]$_2$, ϕ_2很好吃。
My sister visited me. (She) gave me several apples. (They) were delicious.

[1] In the examples, we use ϕ_i to denote a ZA, where the subscript i is the index of its antecedent. The noun phrase attached with the same subscript is the antecedent of the ZA. Below the examples, we also give English translation, in which the unexpressed elements are recovered and identified by ( ).

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(2) 我久睡在一角的[席子]$_3$上, ϕ_3已经烤得那么热。
I was sleeping on a corner of the matting. (It) was heated up very hot.

(3) [这个问题]$_4$看上去很容易, 我们还没有解决ϕ_4。
The problem seems to be very easy, but we have not resolved (it) yet.

ZA resolution is important in many natural language processing tasks. One example is machine translation (MT). For target languages such as English that cannot be adequately generated with omitted expressions, here the subject and object, the antecedent of the ZA in the source language must be identified and made explicit.

Observe the translation outputs for sentence (1) produced by some existing Chinese-English MT systems.

[Reference translation]: My elder sister visited me. (She) gave me several apples. (They) were delicious.

[MT1]: The elder sister comes to see me, gives me several apples, is delicious.

[MT2]: Sister come to see me, gave me a few apples, good to eat.

[MT3]: The elder sister comes to see me, giving me a few apples, delicious.

From these translation outputs, it is obvious that the three MT systems cannot deal with ZA resolution. That is they cannot perform to detect ZAs, identify their antecedents and generate the grammatical English in the further. Besides the systems above, the majority of current MT systems only handle one-sentence inputs, rather than full discourses. Thus they usually cannot solve ZA problem, which should be operated beyond the sentence level [3]. Our study is motivated by improving the quality of MT by resolving ZA problem.

In this paper, we assume that the task of ZA detection has been performed by some other module, such as a shallow parser [4]. Our work only focuses on identifying the antecedent of ZA. We construct a maximum entropy (ME) classifier to check whether a candidate is the correct antecedent or not. In training and applying the classifier, we first employed a set of 13 regular features to capture the context information in discourses. From the original experiment results, we found that the classifier using the regular features mainly suffered from the problem of insufficient semantic knowledge. Therefore, we improved the classifier by developing two Web-based features to obtain additional semantic information from the Web. The values of the two features can be estimated by querying the Web. We combine the two features with the regular feature set and retrain the advanced ME-based classifier. In this way, we effectively incorporate the Web into our classifier and experimentally show that the Web as a knowledge source can improve the performance of our approach. In the application of our ZA resolution in MT, we propose a frame combining our resolution and MT using a detachable and MT-independent pre-processing module. BLEU/NIST measures are used to evaluate the performance of three Chinese-English MT systems without and with the ZA resolution respectively. The experiment results show that the ZA resolution can improve the quality of MT significantly.

2 Previous Research About ZA Resolution

For identifying anaphoric relations, the existing research can be categorized into 3 classes: rule-based, statistical or machine learning models, and hybrid approaches.

In the first class, most approaches have been using heuristic rules to eliminate unacceptable candidates until the most likely antecedent is obtained. These hand-crafted rules were developed based on pragmatics and a set of linguistic factors, which include gender, number agreement, semantic consistency, syntactic parallelism, salience, proximity etc [4] [5] [6]. However, writing a complete rule set is very labor-intensive and time-consuming work, and also requires the linguistic expertise. Furthermore, if we want to take more factors and knowledge into account, it is difficult to maintain the complicated heuristic rules. In contrast to the labor intensive rule-based methods, the second class employs statistical models [7] and machine learning techniques [8]. The methods in this class rely heavily upon the availability of large corpora annotated with anaphoric relations. However, collecting such a training corpus also requires a lot of work. In order to alleviate the problem that a lot of manual work is required, [9] proposed a hybrid approach, which combined heuristic ranking rules and machine learning system, SVM. Their results showed that the combination made Japanese zero pronoun resolution more reliable. A similar hybrid architecture has been employed in [10], which combines a rule-based pre-filtering component with a memory-based resolution module for anaphora resolution in German.

Although the approaches discussed above have made great contributions to ZA resolution, most of them still have shortcomings. First, test data in most experiments were selected from one single genre, such as newspaper articles or technical manuals. The problem of whether the proposed approaches are also effective on other genres is unexplored. Semantic knowledge like common sense, is always ignored in resolution because such kind of knowledge is hard to be captured from contexts. Finally, for Chinese texts, no attempt has been made to employ a machine learning framework.

3 Our Approach for ZA Resolution

3.1 ME-Based Classifier

Maximum Entropy (ME) is a general technique for estimating probability distributions from data. The general idea of ME is that the best probability model for the data is the one that maximizes entropy over the set of probability distributions that are consistent with events [11]. In the ME framework, events are represented as multiples of weighted features. In training, the weights of features are derived on the basis of the distribution of the features in the training data, using an iterative algorithm such as generalized iterative scaling (GIS) or improved iterative scaling (IIS) [12]. In employing the model, events of a given context are evaluated by summing the weights of their respective features and normalizing over the context to obtain a probability distribution, as in Eq. 1.

$$p(y|x) = \frac{1}{Z(x)} exp\left[\sum_i \lambda_i f_i(x,y)\right] \quad (1)$$

where $p(y|x)$ represents the probability of event y given context x, and λ_i represents the weight for feature f_i. $Z(x)$ is simply a normalizing factor to ensure a proper probability.

We develop a classifier based on ME to determine whether a candidate is the correct antecedent or not. Given a candidate $x \in$ {all possible candidates for a certain ZA}, we need to predict $y \in$ {yes, no}, which classifies x as the correct antecedent or not. The classification process starts from the noun phrase (NP) immediately preceding the ZA and proceeds backwards in the reverse order of the discourse until an antecedent is found.

The primary work of modeling an ME-based classifier is the selection of a set of features (f_i) to be considered in the analysis of the training data and the estimation of the appropriate weight (λ_i) for each feature. In the latter work, we use IIS to find the globally optimal weights given the training data. The features devised for ME modeling are described in Sect. 4.

3.2 Experimental Data

Instead of considering the ZAs of only one genre, that is, newspaper articles [3,6], we have used a diverse text corpus, which is a collection of 85 text files from different genres, about 30 million words in all, consisting of newspaper articles (current affairs, politics, economy, sports and entertainment), Chinese fiction written by different authors and textbooks from elementary school to graduate school level. We randomly selected two subsets of the corpus as our training and test data. The data excludes three types of ZA: intra-sentential, exophora and cataphora. The training data consists of 85 fragments, one fragment from each text file, with a total of 19,293 Chinese words. It includes 627 ZAs in all. There are 3,027 words and 121 ZAs in the test data.

Raw text was input and automatically preprocessed by a pipeline of NLP modules, consisting of sentence segmentation, POS tagging, NE recognition, shallow parsing, nested NP extraction and semantic class determination. All possible annotations in a training text are determined by the outputs of these modules. Our POS tagging and NE recognition were done on the basis of the Chinese lexical analysis system ICTCLAS[2]. ICTCLAS is an approach based on multilayer HMMs and it includes word segmentation, POS tagging and NE recognition. Our shallow parser parses a sequence of words into smaller constituents such as NP and verb phrases (VP) with phrase-level parsing. The module of nested NP extraction divides nested NPs into two groups. They are nested NPs that are modifier nouns. For example, the nested NP for "shi yan jie guo (experimental result)" is "shi yan (experiment)", and nested NPs that modify the heads using the auxiliary word "de (of)". For example, for "er tong de jiao yu

[2] http://sewm.pku.edu.cn/QA/reference/ICTCLAS/Free ICTCLAS

(children's education)", the nested NP is "er tong (children)". Semantic classes of phrases are determined based on the Semantic Knowledge-base of Contemporary Chinese (SKCC) [13]. We defined 9 semantic classes for NPs and 7 classes for VPs.

Each ZA was extracted and considered in a 4-sentence context. The correct antecedents were manually annotated. In the training data, each pair of a ZA and its antecedent is considered to be a positive training instance for our classifier, while all the other NPs occurring in the scope between the ZA and it antecedent form negative training instances with the ZA. These NPs consist of general NPs, nested NPs and NEs. This procedure produced a set of 3,261 training instances, in which 627 (19.2%) were positive and 2,634 (80.8%) were negative. In the test data, all the NPs extracted in the context for a certain ZA are regarded as candidates and are used in the later classification experiments.

4 ZA Resolution with the Regular Features and Web-Based Features

4.1 Regular Features

In this section, we describe the ZA resolution using a set of regular features. The values of the features can be obtained directly from the corpus, from the pipeline of NLP modules, e.g., our POS tagger and shallow parser and using semantic dictionaries with the help of heuristic rules. The 13 regular features, RF, includes 4 grammatical, 5 lexical, 1 positional and 3 semantic features (Table 1):

Grammatical features. PREDI-ZA and PREDI-A are the surface forms of the heads of the predicates in the sentences containing the ZA and the antecedent candidate. The values come from the POS tagger and shallow parser. The feature NESTED-A checks whether the antecedent candidate is a nested NP or not. The feature GRAM-FUNC shows the grammatical role of an antecedent candidate and the values are obtained using our shallow parser with some heuristics.

Lexical features. The feature P-P describes the lexical relation between the heads of the predicates in the sentences containing the ZA and the candidate. The four possible values can be obtained based on a synonym/antonym dictionary. The feature DEMON-A checks whether the candidate starts with a demonstrative pronoun or not. PRON-A and NE-A judge whether the candidate is a pronoun or a NE. The feature CONJ evaluates the type of conjunction between the ZA and the candidate. The values can be determined directly by the conjunctions occurring in the sentences.

Positional feature. The feature DIST measures the distance between the antecedent candidate and the ZA in numbers of simple sentences.

Semantic features. The features SEM-A, SEM-P and SEM-P-A describe the semantic classes of the antecedent candidate and the heads of the predicates in the sentences containing the ZA and the candidate, respectively. All the values of these three features can be determined using a Chinese semantic dictionary.

Table 1. Summary of regular features (RFs)

Type	No.	Feature	Description	Values
Grammatical	1	PREDI-ZA	Surface form of the head of the predicate in ZA-S.	VP, AP, PP, MP, NP, unknown
	2	PREDI-A	Surface form of the head of the predicate in A-S.	VP, AP, PP, MP, NP,unknown
	3	NESTED-A	The candidate is a nested NP.	yes, no
	4	GRAM-FUNC	The grammatical role of the candidate.	subject, object, embedded subject, embedded object, attributive, adverbial modifier, others
Lexical	5	P-P	Lexical relation of the heads of the predicates in ZS-S and A-S.	same, synonym, antonym, others
	6	DEMON-A	The candidate starts with a demonstrative pronoun.	yes, no
	7	PRON-A	The candidate is a pronoun.	yes, no
	8	NE-A	The candidate is a NE.	yes, no
	9	CONJ	Type of conjunction between the candidate and the ZA.	coordinate, continued, causality, transitional, others
Positional	10	DIST	Distance between the candidate and ZA in sentences.	1, 2, 3, 4
Semantic	11	SEM-A	Semantic class of the candidate person.	animal, plant, object, artifact, space, time, data, process ...
	12	SEM-P	Semantic class of the head of the predicate in ZA-S.	state, emotion, change, weather, perception, communication ...
	13	SEM-P-A	Semantic class of the head of the predicate in A-S.	state, emotion, change, weather, perception, communication ...

ZA-S: the sentence containing the ZA; A-S: the sentence containing the antecedent candidate
VP: verb phrase; AP: adjective phrase; PP: prepositional phrase; MP: numeral-classifier phrase

4.2 Experiment Results of the Classifier Trained on the Regular Features

In order to show the improvement of the performance of our approach, we developed the following simple rule-based baseline algorithm.

Baseline rule
For each ZA in a sentence, choose the NP in the previous sentence having the shortest distance from the ZA as its antecedent.

Table 2 shows the results for the ME-based classifier using RF in comparison with the baseline on the test data (121 ZAs). The performance is evaluated as to *accuracy*, that is, the number of correctly resolved ZAs divided by the total number of ZAs.

Our approach performs much better than the baseline. While these results are encouraging, there were several classification errors. Except the errors caused by the modules such as POS and NE recognition, the largest type of errors is due to the lack of semantic knowledge (47.1%). The examples having this type of errors can roughly be classified into two groups:

Table 2. Results with RFs

Feature	Baseline	RF
Correct number	56	87
Accuracy	46.3%	71.9%

1. Examples having word senses that are missing from the semantic dictionaries, e.g., "second quarter" should be assigned the semantic class "data" instead of "unknown" in example (4) below.
2. Examples that require general world knowledge beyond simple word meanings, e.g., examples (5) and (6) below. In (5), ϕ_6 and ϕ_7 can be interpreted correctly using the knowledge that bananas, being a fruit, are likely to be unripe and the monkeys, being an animal, are likely to be hungry. In (6), ϕ_8 refers to "Li Si" because "Zhang san" no longer has the apple. The general world knowledge on "give" is that the actor no longer has the object after giving it to others.

(4) 每年的[第二季度]$_5$收入都会减少, ϕ_5一直是PC界最差的销售季度。
Revenue decreases in the second quarter of each year. (The second quarter) is always the slowest quarter of the year for the PC industry.

(5) 他掏出准备好的[香蕉]$_6$, 虽然ϕ_6没熟, 他给了[猴子]$_7$, 因为ϕ_7实在是饿了。
He took out the prepared bananas. Although (they) were not ripe, he gave them to monkeys. Because (the monkeys) were very hungry, they hurried up and scrambled for them immediately.

(6) 张三给[李四]$_8$一个苹果, ϕ_8很快就吃了。
Zhang san gei Li si yi ge ping guo, (Li si) hen kuai jiu chi le.
Zhang San gave Li Si an apple. (Li Si) ate the apple.

From the error analysis above, it is obvious that our approach mainly suffers from semantic problems, such as semantic ambiguity and the lack of semantic knowledge. To our knowledge, these problems cannot be resolved using any current semantic dictionary.

4.3 Web-Based Features

The Web has a vast amount of data and the volume is much larger than that of any normal corpus. Therefore, the Web is increasingly being used as an information resource in a wide range of NLP tasks. Lapata and Keller [14] describe how to use Web-based models for resolving eight NLP tasks and show the promising performance of those unsupervised systems.

In our work, we employ the Web as a semantic knowledge resource on the basis of two facts discovered by careful examination of the experiment examples. One fact in ZA resolution is that the semantic constraint that is satisfied by a ZA, must be satisfied by its antecedent. That means that for a certain ZA, its antecedent and predicate must have semantic consistency. For example, considering ϕ_6 and ϕ_7 in (5), the antecedent of ϕ_6 should be "bananas" because the

candidate "bananas" has higher semantic consistence than "monkeys. The other fact is that although the relation between a ZA and its antecedent is implicitly expressed, they are linked by a strong semantic relation between the heads of the predicates of the sentences containing the ZA and its antecedent, denoted as V_{antec} and V_{ZA}. For example, considering ϕ_8 occurring in the subject position in (6), the head pair [give :: eat] has the *transfer* relation, which indicates two actions are conducted by different subjects, that is, the subject is supposed to change from one sentence to the next. Therefore the antecedent of ϕ_{13} is not the subject of the previous sentence, "Zhang san", but "Li Si". Examples of head pairs of the *transfer* relation include [show :: see], [fell :: fall], and [give :: have]. Besides the *transfer* relation, another relation, the *happens-before* relation defined between V_{antec} and V_{ZA} is considered in our work. This type of relation indicates that the two disjointed actions are conducted by the same subject. Examples of this type of relation include [marry :: divorce], [enroll :: graduate], and [get up :: dress].

From the points discussed above, it is obvious that the semantic consistency and the semantic relations between predicates could benefit ZA resolution. We compute the semantic consistency and discover the semantic relations by querying the Web with certain patterns indicative of each relation. We extract the Web knowledge by the following procedure.

1. **Construct lexico-syntactic patterns.** To compute the semantic consistency, two kinds of predicate-argument relations, subject-predicate and predicate-object, are considered in our work. The patterns are constructed in the form of "NP VP" (for subject-predicate) and "VP NP" (for predicate-object). The patterns for discovering the semantic relations between V_{candi} and V_{ZA} are manually designed and refined by examining examples in a volume of known semantic relations. For example, to detect the *happens-before* relation, the patterns used could be "xian V_{candi} zai V_{ZA} (V_{candi} and then V_{ZA})" and "V_{candi} jie zhe V_{ZA} (V_{candi} and later V_{ZA})". To detect the *transfer* relation, the patterns could be "bei V_{candi} ran hou V_{ZA} (be V_{candi}ed and then V_{ZA})" and "bei V_{candi} jie zhe V_{ZA} (be V_{candi}ed and subsequently V_{ZA})". In our work, we use a total of 17 patterns for extracting semantic knowledge from the Web.
2. **Instantiate patterns.** We instantiate the patterns for all antecedent candidates. For example, for ϕ_6 in (5), the pattern "NP VP" is instantiated with "the bananas are ripe" and "the monkeys are ripe". If a candidate is a pronoun, it would be replaced by its closest nominal antecedent before instantiation. These instantiated patterns are submitted to a Web search engine (Google in our work).
3. **Acquire semantic knowledge from the Web.** To compute the semantic consistency from the Web, our rationale is that if the consistency between the ZA and a candidate is high, it is likely that the pattern instantiated by the candidate occurs frequently on the Web. We use *ConsiScore* to estimate semantic consistence using Eq. (2), where *count(P)* is the Web count of the instantiated pattern returned by Google, while *count(candi)* is the Web

count of the query formed with only the head of the current candidate. N is the number of Google pages ($N \approx 10.1 \times 10^9$). To determine the semantic relations, we adopt a method inspired by mutual information (MI) to measure the strength of association between V_{candi} and V_{ZA}, denoted $R_P(V_{candi}, V_{ZA})$, which is estimated using Eq. (3), where $count(V_{candi})$ and $count(V_{ZA})$ are the Web counts of the queries formed by V_{candi} and V_{ZA}. We determine that the semantic relation R_P indicated by pattern P is present between V_{candi} and V_{ZA} if $R_P(V_{candi}, V_{ZA})$, exceed a certain threshold θ (θ=0.3).

$$ConsiScore(candi, ZA) = \frac{count(P)}{count(candi)} \tag{2}$$

$$Rp(V_{candi}, V_{ZA}) = \frac{count(P) \times N}{count(V_{candi}) \times count(V_{ZA})} \tag{3}$$

Unlike research in which web counts are used directly [14], our method of employing the semantic knowledge from the Web is to combine the Web knowledge as features together with the 13 RFs described in Sect. 4.1. The two Web-based features with their possible values are summarized in Table 3. The feature SEM-CONSI obtains the semantic consistency computed from the Web. The values represent the rank of the consistency. For example, SEM-CONSI=1 means the current candidate has the highest semantic consistency among all the antecedent candidates. The feature SEM-RELA represents the semantic relations between the heads of the predicates of ZA and a candidate, as described in Sect. 5.1. In our current work, three values are possible: "transfer", "happens-before" and "unknown".

Table 3. Two Web-based features

Type	No.	Feature	Describe	Values
Web-based	14	SEM-CONSI	The rank of the semantic consistency of antecedent candidates.	1, 2, 3, 4 ...
	15	SEM-RELA	The semantic relations between the heads of the predicates in sentences containing ZA and an antecedent candidate.	transfer, happens-before, unknown

4.4 Experiment Results on the Regular Features and the Web-Based Features

We retrained and tested the advanced classifier with the RF set, and the two additional Web-based features. The results are summarized in Table 4. The rightmost column shows the performance of the advanced classifier. We obtain 35.5 and 9.9 point improvements in accuracy in comparison with the baseline and the original classifier trained on the RF set. In particular, all the examples given in this paper can be resolved by the advanced classifier.

Table 4. Results with RF and RF+Web-based features

Feature	Baseline	RF	RF+Web
Correct number	56	87	99
Accuracy	46.3%	71.9%	81.8%

Table 5. Personal pronouns table

	Singular	Plural
1st person	我 (I)	我们(we)
2nd person	你 (you)	你们(you)
3rd person	他, 她, 它(he, she, it)	他们, 她们, 它们 (they, they, they)

5 Incorporating ZA Resolution into MT

5.1 The Frame of Incorporating ZA Resolution into MT

We incorporate the ZA resolution into MT using a general type of automated pre-processing module defined. The advantage of general modules is they are detachable and independent of any particular MT system. Fig. 1 shows the processing flow of applying ZA resolution in MT.

In Fig. 1, for the input of Chinese text, ZA resolution can be viewed as pre-processing of the input, in which ZAs are detected and their antecedents are identified. We replace ZAs by the pronouns denoting their antecedents based on personal pronouns table (Table 5), and then input the corrected texts into MT systems.

For example, Suppose the sentence to be translated is sentence (1) in Section 1. After ϕ_1 and ϕ_2 are recovered by "她 (she)" and "它们 respectively, the corrected sentence becomes to be (1a). (MT1), (MT2) and (MT3) are the translation outputs for sentence (1a) produced by three existing Chinese-English MT systems.

(1a)[姐姐]$_1$来看我, (她)给了我几个苹果, (它们)很好吃。

[Reference translation]: My elder sister visited me. (She) gave me several apples. (They) were delicious.

Table 6. The experiment results of MT evaluation

MT	BLEU (4-gram)		NIST (4-gram)	
	without ZA Resolution	with ZA Resolution	without ZA Resolution	with ZA Resolution
Babel Fish	0.462	0.619	5.198	6.807
Google	0.310	0.413	4.201	5.635
Kingsoft	0.509	0.622	5.704	7.113

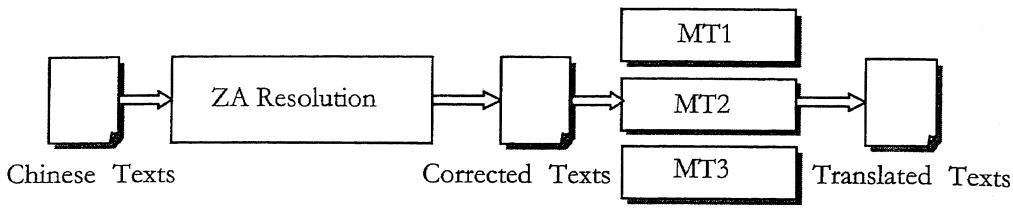


Fig. 1. The processing flow of applying ZA resolution in MT

[MT1]: The elder sister comes to see me, she gives me several apples, they are delicious.
[MT2]: Sister come to see me, she gave me a few apples, they are good to eat.
[MT3]: The elder sister comes to see me, she gives me a few apples, they are delicious.

5.2 The Performance of MT with ZA Resolution

We collect 200 pairs of aligned Chinese-English bilingual sentences from Chinese Linguistic Data consortium as the test data. There is at least one ZA in each of the Chinese sentences. The English sentences are viewed as reference translations in later evaluation experiments. Three existing Chinese-English MT systems are investigated in our evaluation. They are Babel Fish MT[3] , Google MT[4] that are free translation services online, and Kingsoft MT[5], which is one of the most famous translation software used in China.

The automated evaluation of MT is carried through by using reference translations for matching the translated texts. There are the N-gram co-occurrence evaluations that analyze the agreement of terms and their sequences (N-gram) between the evaluated and reference translations. In our experiments, we use BLEU and NIST as the evaluation metrics. BLEU averages the precision for unigram, bigram and up to 4-grams and incorporates a size penalty [15]. The NIST metric is derived from the BLEU evaluation criterion but differs in one fundamental aspect. That is instead of n-gram precision the information gain from each n-gram is taken into account.

The BLEU and NIST scores of the 3 MT systems without and with ZA resolution are summarized in Table 6. For Babel Fish MT, BLEU and NIST scores increase from 0.462 to 0.619 and from 5.198 to 6.807. For Google MT, the scores increase from 0.310 to 0.413 and from 4.201 to 5.635. And for Kingsoft, they increase from 0.509 to 0.622 and from 5.704 to 7.113. A significant improvement on evaluation scores after the ZA resolution incorporation into the 3 MT systems is demonstrated.

[3] Babel Fish Translation: http://babelfish.altavista.com/
[4] Google Translation: http://translate.google.com/
[5] Kingsoft: http://www.kingsoft.net

6 Conclusion

In this paper, we presented a machine learning approach based on ME for ZA resolution in Chinese. Besides commonly used features, we developed two innovative Web-based features for additional semantic knowledge from the Web. Our approach shows that the semantic knowledge obtained from the Web can be effectively introduced into the machine learning framework and significantly improve the performance of the ZA resolution. In the application in MT, ZA resolution is viewed as a pre-processing module to correct the Chinese texts that have ZAs. The BLEU/NIST scores demonstrate a significant improvement after the ZA resolution incorporation in different kinds of MT systems.

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